SANJAY RAM SRINIVASAN

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Professional summary

AI/ML Engineer and Computer Science Engineering student specialising in Artificial Intelligence and Machine Learning at VIT Andhra Pradesh. Experienced in designing and deploying Retrieval-Augmented Generation (RAG) systems, Large Language Model (LLM) integrations, deep learning pipelines, and AI-powered developer tools. Proficient in Python, FastAPI, LangChain, LangGraph, TensorFlow, PyTorch, Scikit-learn, OpenCV, Natural Language Processing (NLP), Computer Vision, and vector databases. Strong foundation in Machine Learning, Deep Learning, Generative AI, Data Structures & Algorithms, Object-Oriented Programming, and Database Management Systems, with experience building scalable AI and backend applications.

Technical skills

Languages: Python, Java, JavaScript, SQL
AI & Machine Learning: Machine Learning, Deep Learning, Neural Networks, Large Language Models (LLMs), Natural Language Processing (NLP), Computer Vision, Generative AI, Scikit-learn, TensorFlow, PyTorch, HuggingFace Transformers, NumPy, Pandas, OpenCV
Generative AI: Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, Embeddings, Semantic Search, Vector Databases
Data & Analytics: Data Analysis, Data Science, Model Training, Model Evaluation, Model Deployment
Backend Development: FastAPI, Flask, SQLAlchemy, REST APIs, API Development, Pydantic
Frontend: HTML5, CSS3, Bootstrap, React
Databases: MySQL, SQLite, ChromaDB, Vector Databases
Cloud & DevOps: AWS, Docker, Git, GitHub, CI/CD, Render
Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks
Soft Skills: Problem Solving, Analytical Thinking, Collaboration, Communication, Attention to Detail
Developer Tools: VS Code, Jupyter Notebook, Postman, Figma

Experience

Software Development Intern | Identity World **Jun. 2025 – Jul. 2025**
Bengaluru, Karnataka, India

- Developed a seating arrangement solution for a hackathon to organize participant seating and improve event coordination.
- Designed seating allocation logic to assign participants to appropriate seats based on participant information, seating capacity, and event requirements.
- Implemented a structured approach for managing participant and seating information to simplify arrangement and coordination.
- Translated event-management requirements into a functional software solution while working within a real-world hackathon environment.

Projects

Self-Healing RAG Platform | Enterprise AI Knowledge Assistant **Jun. 2026 – Jul. 2026**
Tech: Python · FastAPI · LangChain · LangGraph · ChromaDB · Groq API · React

- Developed an enterprise **Generative AI and Retrieval-Augmented Generation (RAG)** platform enabling natural language interaction with PDF, DOCX, and text documents through LLM-powered semantic search.
- Built an end-to-end **document ingestion pipeline** for document processing, chunking, embeddings generation, vector indexing, and context retrieval using ChromaDB.
- Designed a **self-healing retrieval workflow** using LangGraph that validates retrieved context, rewrites ambiguous queries, retries failed retrievals, and refines responses through iterative self-correction.

- Integrated **FastAPI REST APIs** with LangChain, ChromaDB, and Groq LLM inference to generate context-aware, citation-backed responses grounded in user-provided documents.
- Implemented an analytics dashboard monitoring **confidence scores, retrieval quality, hallucination rate, and answer relevance** for RAG system evaluation.

Prometra | AI Developer Intelligence Platform **Jul. 2026 – Aug. 2026**
Tech: Python · Typer · SQLite · Pydantic · Git · Rich · Pytest

- Built a local-first **Python AI Developer Intelligence Platform** for tracking coding sessions, filesystem activity, Git history, and project evolution while maintaining local data storage.
- Developed modular **Filesystem, Git, Timeline, Replay, Search, Diff, and Compare engines** to capture and analyze end-to-end software development workflows.
- Designed a Typer-based **command-line interface (CLI)** with 15+ commands for project initialization, session management, timeline visualization, analytics, search, replay, and report generation.
- Implemented **AI Session Intelligence** using NLP techniques to generate coding summaries, productivity analytics, coding pattern analysis, and developer health scores from project activity.
- Generated multi-format **Markdown, HTML, JSON, and CSV reports** for analyzing development history, Git timelines, and productivity metrics.
- Maintained a CI-ready testing pipeline with **130+ automated tests** across unit, integration, and edge-case scenarios to improve reliability and maintainability.

Early Alzheimer's Disease Detection using Deep Learning **Jul. 2026 – Present**
Tech: Python · TensorFlow · OpenCV · FastAPI · NumPy · Pandas

- Developed a **deep learning and computer vision** system for early Alzheimer's disease detection using MRI brain scan datasets and TensorFlow.
- Built an end-to-end **machine learning pipeline** covering MRI preprocessing, image augmentation, dataset preparation, model training, evaluation, and inference.
- Applied computer vision techniques using **OpenCV** for image preprocessing, normalization, augmentation, and feature extraction to improve model input quality.
- Developed **FastAPI REST API endpoints** for ML model inference with structured JSON responses, enabling integration with web-based applications.
- Designed a modular architecture supporting future **model retraining, dataset versioning, deployment, and scalable AI-powered healthcare applications**.

Education

Vellore Institute of Technology — Andhra Pradesh **Sep. 2023 – Present**
Bachelor of Technology (B.Tech.) in Computer Science & Engineering | AI & Machine Learning Specialisation | CGPA: 7.62

- **Relevant Coursework:** Data Structures & Algorithms · Object-Oriented Programming · Database Management Systems · Computer Networks · Machine Learning · Artificial Intelligence

Sri Chaitanya Techno School **Jun. 2022 – May 2023**
Intermediate — Class XII, CBSE Board